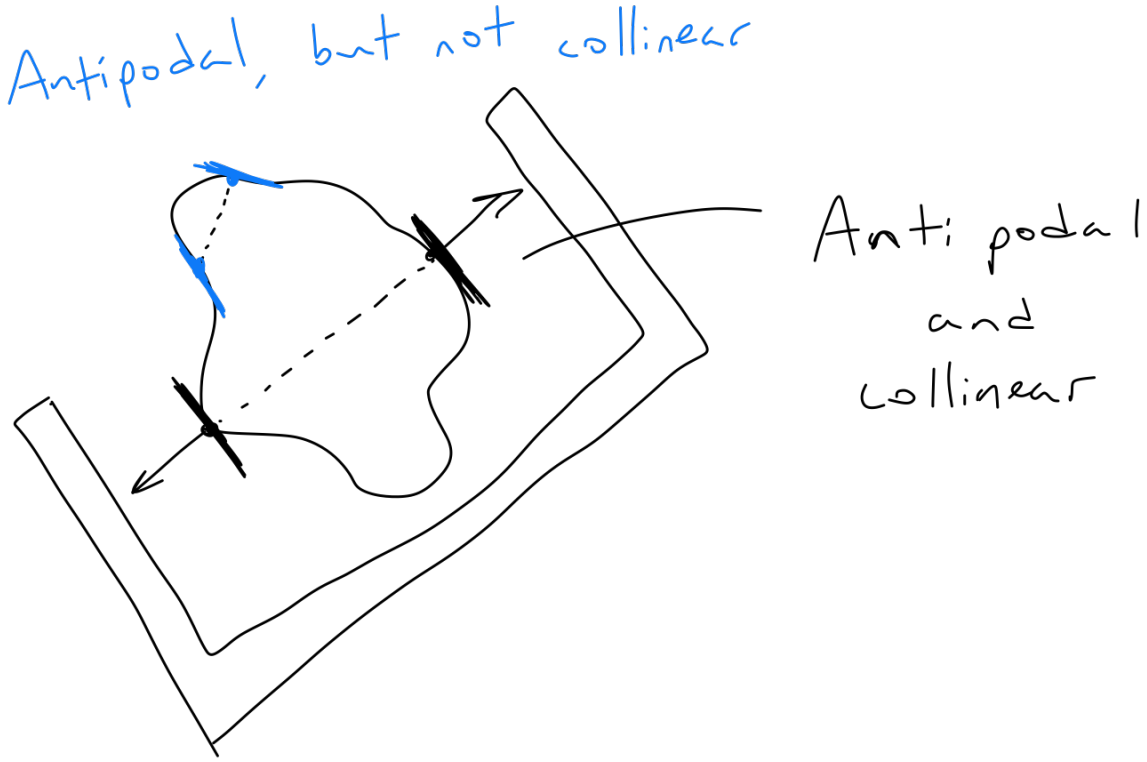
# Geometric grasp selection




## Basic strategy

- Obtain a point cloud (clean it up) compute surface normals for each point.
- Identify antipodal (opposite sides) and collinear (normals align) grasps.
- Avoid collisions, respect joint angles etc.

# Point clouds

- Clean up point cloud
  - Crop
  - Color segmentation
  - Image segmentation (possibly from a learned segmentation model)
- Estimate surface normals for each point
- Merge point clouds from separate cameras as needed (e.g. ICP)
- Down sample

(this order is specific)

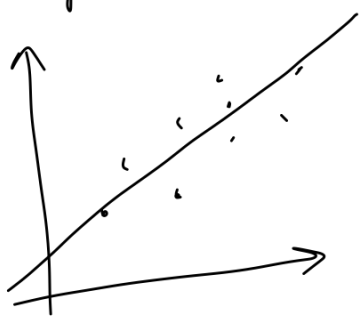
# Normal Estimation

For every point:

- K nearest neighbors (with distance limit)
- Fit a plane to local point cloud
- Flip normals to point at camera

How should we fit a plane to a set of points

Example



In 2D,
find the best fit line.

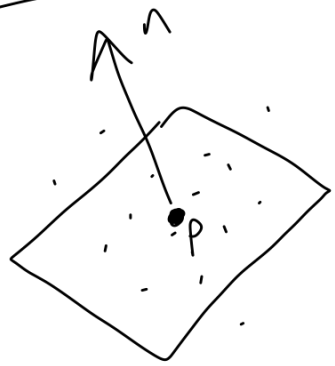
Some options

Least square method

PCA or SVD

with 1 component

In 3D




PCA with 2 components.

The 2 components define the best-fit plane.

normal $n$ + point $p$ in plane

# Down Sampling

- Fill the space with voxels (cubes)

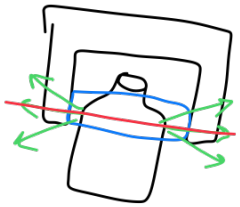
For each voxel:

If voxel contains any points:

- Compute voxel point as (1) middle of the voxel or (2) centroid of all points in the voxel.

- Average all the normals.

# Picking antipodal grasps



Region of points between fingers

Surface normals of points

Line between gripper fingers

## Procedure

Move point cloud to gripper frame

Crop points to between the fingers

Reward points that contribute a normal close to the gripper line, based on a dot product.